

Coastal Carbon Corridor: Mangrove Restoration and Coastal Greenbelt Protection in the East Coast of Aceh and North Sumatra Province, Indonesia



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Document prepared by Yagasu Aceh and Agresta S. Coop

Project Title	COASTAL CARBON CORRIDOR: Mangrove restoration and coastal greenbelt protection in the East Coast Of Aceh And North Sumatra Province, Indonesia.
Version	3
Report ID	001
Date of Issue	20-November-2019
Project ID	1493
Monitoring Period	01-10-2015 – 31-07-2019 for VCS
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Table of Contents

1.	Project Details	4
1.1.	Summary Description of the Implementation Status of the Project	4
1.2.	Sectoral Scope and Project Type.....	5
1.3.	Project Proponent.....	5
1.4.	Other Entities Involved in the Project.....	6
1.5.	Project Start Date.....	6
1.6.	Project Crediting Period	6
1.7.	Project Location	6
1.8.	Title and Reference of Methodology	9
1.9.	Other Programs.....	10
1.10.	Sustainable Development	10
2.	Implementation Status.....	11
2.1.	Implementation Status of the Project Activity	11
2.2.	Deviations	12
2.2.1.	Methodology Deviations	12
2.2.2.	Project Description Deviations	12
2.3.	Grouped Project.....	13
2.4.	Safeguards.....	15
2.4.1.	No Net Harm	15
2.4.2.	Local Stakeholder Consultation	15
3.	Data and Parameters	16
3.1.	Data and Parameters Available at Validation	16
3.2.	Data and Parameters Monitored	26
3.3.	Monitoring Plan	33
i.	Organizational structure	33
ii.	Stratifying.....	34
iii.	Inventory design	39
iv.	Monitoring of net anthropogenic GHG removals.....	41
4.	Quantification of GHG Emission Reductions and Removals	45

4.1.	Baseline Emissions	45
4.2.	Project Emissions	48
4.3.	Leakage	59
4.4.	Net GHG Emission Reductions and Removals.....	59

1. PROJECT DETAILS

1.1. Summary Description of the Implementation Status of the Project

The Coastal Carbon Corridor (CCC) grouped project objectives include increasing the environmental carrying capacity of mangrove ecosystems on the east coast of Northern Sumatra for climate change mitigation and adaptation, natural disaster risk reduction, biodiversity conservation and local livelihood improvement.

The project region has up to 456,896 potential hectares for plantation and mangrove restoration and protection. The grouped project has started with the implementation of the first project activity instance, which has been implemented as described in the PD.

In the first verification of the project Yagasu included 4,999.98 ha of degraded mangrove. It is important to mention that, regarding the spatial data provided in shapefile format, in the first verification the area included was 5068.69, Some areas have been removed from the first verification to this verification after doing a GIS analysis based on historical images from Google Earth. So, it has been considered to remove those lands that have changed the use of the land (buildings, fishponds, roads, industries) and those lands that seems to be submerged most part of the year and that have not been succeed after the plantation period.

The results of this analysis has given a percentage of land use changed under 1% and of submerged plots under the 2,7% of the project areas. After this work of reviewing the limits of the Project area using GIS analysis, the total area is 4,949.02 ha for the first project activity instance

Also in this second verification 278.23 ha has been included as a new project activity instance. The total area included in this verification is 5,227.25 ha.

Table 1 Planting plant of the first and second project activity instance

Year of plantation/ restoration start	Areas of 1 st project activity instance (ha)	Areas of 2 nd project activity instance (ha)	Area included in the project (ha)
2011	191.18	7.74	198.92
2012	851.32	51.95	903.28
2013	1,986.53	12.73	1,999.26
2014	1,623.19	169.90	1,793.08
2015	296.80	1.34	298.15
2016		26.35	26.35
2017		8.21	8.21
Total project	4,949.02	278.23	5,227.25

The number of GHG credits eligible to be issued as VCUs for the first project activity instance of the first monitoring period between 1-10-2011 to 30-09-2015 was 125,391 t CO₂-e.

The number of GHG credits eligible to be issued as VCUs for the first and second project activity instance of the second monitoring period 1-10-2015 to 30-07-2019 is 271,966t CO₂-e.

1.2. Sectoral Scope and Project Type

The project has been developed under VCS sectoral scope 14 (Agriculture, Forestry, Land Use) and is categorized as the VCS AFOLU category ARR: Afforestation, Reforestation and Revegetation. The project is a grouped project.

1.3. Project Proponent

The project proponents (PP) are Livelihoods Fund and Yagasu Aceh. Livelihoods Fund is an organization supported by private companies who believe in working and learning together with project developers to effect change that promotes sustainable farming practices that can increase food production while preserving our natural resources. Yagasu Aceh is the implementer and developer of the community and biodiversity activities.

Livelihoods Fund was the project proponent when the verification of the grouped project under VCS took place. However, Livelihoods recognizes Yagasu as the implementer and developer of the community and biodiversity activities, raising awareness, securing more funding for these activities to obtain CCB certification. Therefore, Yagasu is considered the project proponent, in addition to Livelihoods Fund¹.

Organization name	Livelihoods Fund
Role in the project	Project developer
Contact person	Mr. Stéphane Perrier,
Title	Technical & Innovation Director
Address	48-50 rue de la Victoria, 75009, Paris, France
Telephone	+33 1 44 35 20 84
Email	sperrier@livelihoods-venture.com

Organization name	Yagasu Aceh
Role in the project	Project developer & implementer / CCB data & documentation

¹ Legal contract between Yagasu and Livelihoods Fund "VCS_Accession-Representation YAGASU 2019" available as supporting documentation,

Contact person	Bambang Suprayogi
Title	CEO of Yagasu
Address	Jalan Sei Kuala No 25 Medan, 20154, Indonesia
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Email	bambang@yagasu.org

1.4. Other Entities Involved in the Project

Organization name	Agresta S. Coop.
Role in the project	Preparation of documentation and advise during the verification process
Contact person	Mrs. María Jesús Toledo
Title	Director of Climate Change Department
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1.5. Project Start Date

The start date of the project activity is June 14th, 2011, which corresponds to the date of the first project activity on ground planting campaign.

1.6. Project Crediting Period

Project crediting period: 14-06-2011 – 13-06-31.

The project crediting period for this grouped project shall be 20 years renewable, beginning on June, 2011 and ending on June, 2031. The actual project longevity will be greater than the crediting period; this is because the project design does not include harvesting or thinning in the project scenario, being wetlands restoration the main goal of the project.

1.7. Project Location

The project sites stretch along 497 km from Ujung Pancu village in Aceh province to Pante Labu in North Sumatra province, crossing 13 districts: Aceh Besar, Banda Aceh, Pindie, Pidie Jaya, Bireun, Lhoksemawe, Aceh Utara, Aceh Timur, Langsa, Aceh Tamiang, Langkat, Medan and Deli

Serdang. The width of the mangrove cover is from 20 m to more than 5 km from the shore. Along this width, the areas are dissected by many coastal streams.

- Zone I & II in Aceh Besar, Banda Aceh, Aceh Utara, Pidie, Pidie Jaya and Bireun. This area follows the northern tip of Sumatra Island. Around 40 km of shoreline was severely damaged by the 2004 Asian Tsunami, with entire communities damaged. Prior to the tsunami, the uses in the area was fishponds or nearby coastal agriculture.
- Zone III in Aceh Timur and Langsa. As with the other zones, this area was historically devoted to aquaculture but nowadays it has been abandoned. It consists of a mosaic of shoreline, interior and island areas. Ongoing pressures include the conversion to oil palm plantation. Priority has given to rehabilitating Pulau Sembilan, an island system with abandoned ponds that are also close to the shoreline. A high concentration of the potential plantation plots is localized in this area.
- Zone IV in Tamiang and Langkat. This area is close to the Karang Gading and Langkat Timur Laut Wildlife Reserve. The entire habitat has been degraded since the 1960s and 1970s; the Reserve has been recovered during the past 25 years, but the nearby unprotected areas remain under pressure. The area reforested is located in the interior of abandoned ponds and along estuary embankments. A high concentration of the potential plantation plots is localized in this area.
- Zone V in Deli Serdang. This area forms a long accretion zone along about 20 km of beaches and estuaries.

Figure 1. Map of Project Zonation (Source: Prepared by the authors. AGRESTA S. COOP.)

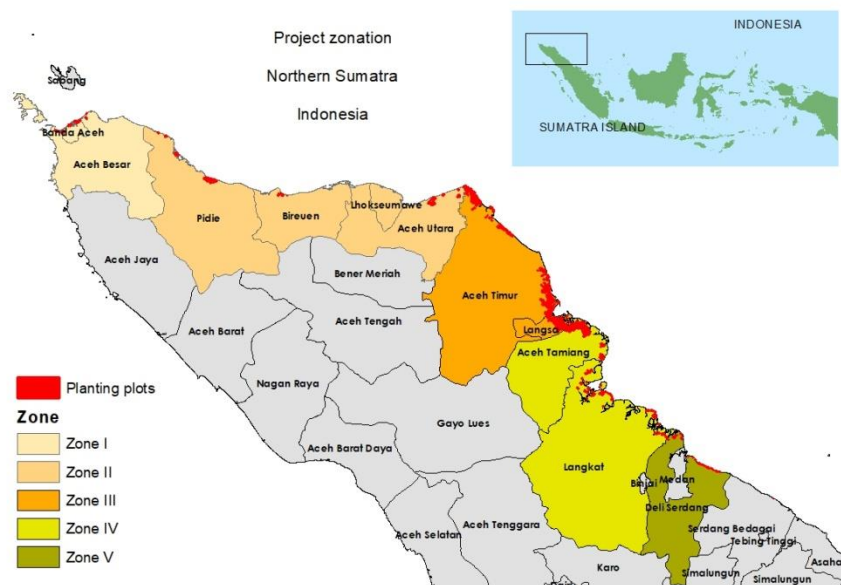


Figure 2. Project boundary of the project (Source: Project Description of the project)

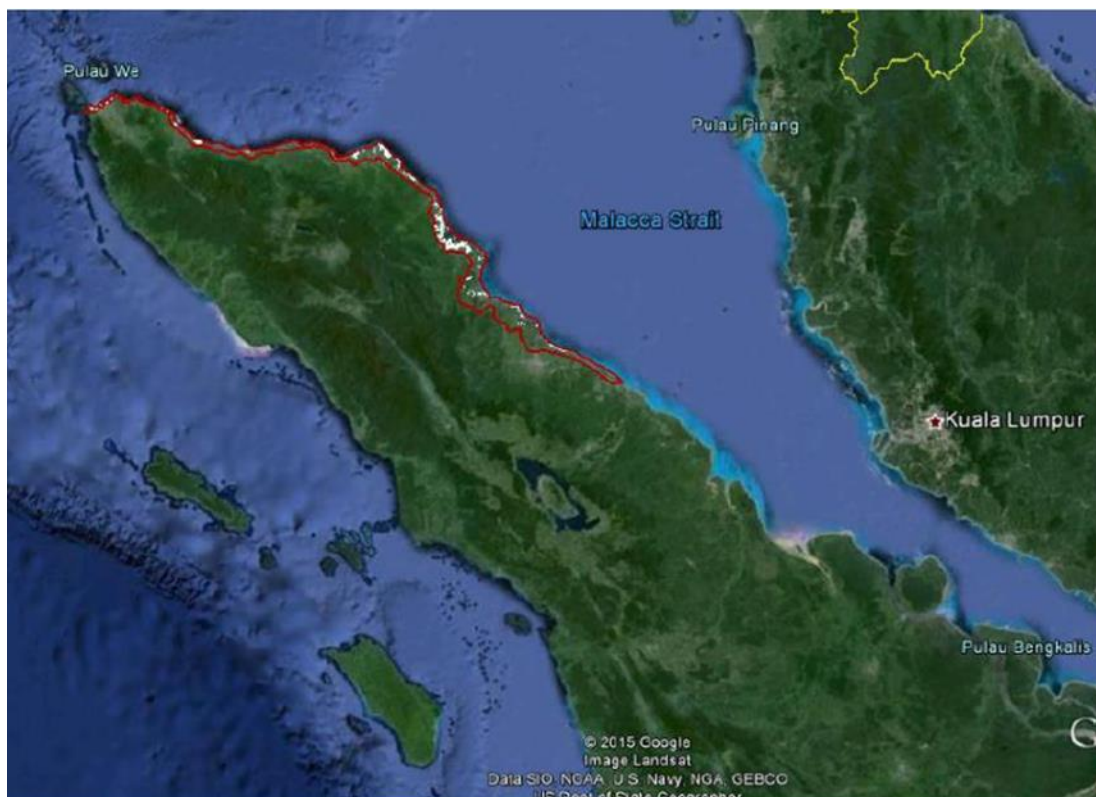


Table 2 Location of the activity project instances

	First activity project instance		Second activity project instance	
	Area (ha)	Number of plantation plots	Area (ha)	Number of plantation plots
Zone I	172.54	253	10.78	50
Aceh Besar	108.57	158	4.30	21
Banda Aceh	63.97	95	6.49	29
Zone II	674.56	1,216	0.86	4
Aceh Utara	203.19	341		
Bireun	17.32	40		
Pidie	74.93	119		
Pidie Jaya	379.12	716	0.86	4
Zone III	2,655.78	3,099	45.76	56

	First activity project instance		Second activity project instance	
	Area (ha)	Number of plantation plots	Area (ha)	Number of plantation plots
Aceh Tamiang	1.58	4		
Aceh Timur	2,095.23	2,426		
Langsa	558.98	669	45.76	56
Zone IV	941.54	1,073	77.12	110
Aceh Tamiang	941.54	1,073	26.71	58
Tamiang			50.40	52
Zone V	504.60	419	143.70	200
Deli Serdang	62.64	50	50.79	73
Langkat	441.96	369	92.49	125
Serdang Bedagai			0.42	2
Total general	4,949.02	6,060	278.23	420

1.8. Title and Reference of Methodology

The project applies the following methodologies:

- AR-AM0014: Afforestation and reforestation of degraded mangrove habitats (Version 3.02).

ARR methodological tools:

- “Combined tool to identify the baseline scenario and demonstrate additionality in A/R CDM project activities” (Version 01)
- Estimation of non-CO2 GHG emissions resulting from burning of biomass attributable to an A/R CDM project activity” (Version 04.0)
- “Estimation of carbon stocks and change in carbon stocks in dead wood and litter in A/R CDM project activities” (Version 03.1)
- “Estimation of carbon stocks and change in carbon stocks of trees and shrubs in A/R CDM project activities” (Version 04.2)
- “Estimation of the increase in GHG emissions attributable to displacement of pre-project agricultural activities in A/R CDM project activity” (Version 02.0)

- Other methodological ARR CDM tools which are applied include, Demonstrating appropriateness of allometric equations for estimation of aboveground tree biomass in ARR CDM project activities (Version 01.0.0)
- Calculation of the number of sample plots for measurements within ARR CDM project activities (Version 2.1.0)

1.9. Other Programs

This grouped project is not participating under other GHG programs. The project is seeking for CCB certification at a later stage after this second VCS verification.

1.10. Sustainable Development

This CCC program² addresses the Indonesia Biodiversity Strategy and Action Plan³ (IBSAP) for implementing national development in accordance with the Conversion of Biological Diversity and strongly links to the National Action Plan for Greenhouse Gas Reduction⁴ (RAN-GRK guidance). More specifically, this initiative is in full alignment with the IBSAP's vision (V) on biodiversity management ("Indonesian biodiversity preservation and development that contributes to national competitiveness and a fair and sustainable use of resources to improve the welfare of current and future generations").

This program also contributes towards the Sustainable Development Goals (SDG):

- SDG01: Economic opportunities will improve the resilience of beneficiaries and increase their net income;
- SDG02: Aquaculture will become a source of sustainable seafood, protein and B-group vitamins in local markets, and thus help reduce food insecurity and local prevalence of malnutrition;
- SDG05: 3 women-led interventions on organic "batik", mangrove foods, and shrimp paste will bring additional income and empower women economically. Overall male/female ratio of beneficiaries: 40/60.

² Coastal Carbon Corridor: <https://www.yagasu.or.id/?p=49>

³ Indonesia Biodiversity Strategy and Action Plan <https://balaikliringkehati.menlhk.go.id/wp-content/uploads/IBSAP%202003.pdf>

⁴ National Action Plan for Greenhouse Gas Reduction:
http://ranradgrk.bappenas.go.id/rangrk/admincms/downloads/publications/Gudeline_for_implementing_green_house_gas_emission_reduction_action_plan.pdf

- SDG13&14: The mangrove restoration and protection will help to restore depleted fish stocks in surrounding marine ecosystems and to sequester over 10m tCO₂e over 20 years. Additionally, this intervention will contribute towards SDG04, SDG12, SDG15 and SDG17.

The program components are already matched with the basic principles of the province's economic development⁵: local resources and environment, community-based, value added, and sustainable market-based approach. The scheme for creating added value for the community's key commodities is to establish an integrated social business model.

Indonesia has been a party to the UNFCCC (United Nations Framework Convention on Climate Change) since August 1994. The country has ratified the UNFCCC (1995), the Kyoto Protocol (2004), and has signed the Paris Agreement (Paris 2015, COP21).

The Republic of Indonesia participates voluntarily in the Clean Development Mechanism (CDM).

2. IMPLEMENTATION STATUS

2.1. Implementation Status of the Project Activity

The grouped project has started with the implementation of the first project activity instance (4999.98 ha), which has been implemented as described in the PD. In the second monitoring period the second project activity instance has been added. As it has been explained in section 1.1 some areas have been removed from the project. Also a second project activity instance has been added. Total area included in this verification period is 5,227.25 ha.

Both the monitoring report and the Non-Permanence Risk assessment report have also been prepared simultaneously with the MR, assuring congruence among the different documents.

The implementing partner of this project has been Yagasu Aceh, a local NGO, and technologies/procedures described in the PD have been followed:

- Logistics organization: organization of teams, headquarters, field bases, means of transport and all the necessary logistics implementation to develop the project
- Training and awareness: local people awareness and training of Yagasu's team have been key issues for the correct implementation of the project. Several activities were designed to motivate local people and make them conscious of the potential positive impacts of mangrove restoration.

⁵ Master Plan Acceleration and Expansion of Indonesia Economic Development 2011-2025, available as supporting documentation.

- Site selection: this activity consists in the identification of the potential project areas to be reforested. This step was performed by the Yagasu's team with the support of scientific experts to identify areas with good ecological conditions for the project, and thanks to the knowledge of the local people of the deforested mangrove areas in the surroundings of their villages.
- Species selection and seedling preparation. There has not been any aerial site preparation in this reforestation project activity. The plantation has been realized manually and has consisted on preparing a small hole for the roots of the seedling, respecting the complete structure of the soil.
- Planting (site preparation, planting, etc.): this activity was performed manually by local communities and with the support of Yagasu based on the technical guide for mangrove reforestation developed by the project with the advice of mangrove experts. Most of the afforestation activities in YAGASU are in abandoned fish ponds and some on coastal/river mudflats. Many of the planting sites were bare land with very few and scattered existing mangrove vegetation below the definition of forest in Indonesia, i.e. minimum area 0.25 ha, minimum crown cover 30%, and minimum tree height 5 m (Yagasu has a SOPs 'Planting mangrove', available as supporting documentation):
- Maintenance: once project areas have been planted maintenance activities are necessary to assure the objective of the project. Maintenance activities have been done by local communities and Yagasu since the starting date of the project.

As has been mentioned previously in this document, a change in the Project Proponent has taken place in this monitoring period. Livelihoods Fund was the project proponent for the first verification of the project. For this verification Livelihoods and Yagasu are both the Project Proponent. Livelihoods has recognized Yagasu as the implementer and developer of the community and biodiversity activities that have been developed inside the Project boundary. The project will probably look for a CCB certification in a later step.

2.2. Deviations

2.2.1. Methodology Deviations

No methodology deviations were made during this monitoring period.

2.2.2. Project Description Deviations

A change in the project proponent has been made in this monitoring period. Livelihoods Fund was the project proponent in the first verification of the project under VCS. However, Livelihoods recognizes Yagasu as the implementer and developer of the community and biodiversity activities, raising awareness and securing more funding for these activities to obtain CCB

certification. Therefore, Yagasu is considered the project proponent, in addition to Livelihoods Fund. A project contract between Yagasu and Livelihoods has been established⁶.

According to VCS Standard rule 3.6.1, deviations from the project description are permitted at verification. A change in the project proponent is a deviation that does not impact the applicability of the methodology, additionality or the appropriateness of the baseline scenario, and the project remains in compliance with the applied methodology. However it shall be described in all subsequent monitoring reports.

Mention should also be made of the change in the title of the project in this monitoring period. "Coastal Carbon Corridor" has been added to the original title "Mangrove Restoration and Coastal Greenbelt Protection in the East Coast of Aceh and North Sumatra Province, Indonesia". The grouped project has been extended into a long-term program named Coastal Carbon Corridor that consists of multi-project activities.

As it has explained before, some of the areas have been removed from the previous verification based on an analysis with historical images from Google Earth. There have been some cases of encroachment in this monitoring period. The land use changes have affected 32.43 ha, meaning, less than 1% from total area. Nevertheless, since there is no likelihood of these polygons or part of polygons being reforested there is no point in continuing to monitor and thus these areas have been removed from the project area. Also some polygons have been removed for this verification because they seem to be submerged most part of the year and that have not been succeed after the plantation period. The land removed classified as submerged plots sum up to 87.24 ha, what means about the 2,7% of the project area⁷. In total 119.67 ha have been removed from previous verification.

These changes from the validated PD have occurred during this monitoring period. It has been determined that these deviations do not impact the applicability of methodology, additionality or the appropriateness of the baseline scenario. The project remains in compliance with the applied methodology.

2.3. Grouped Project

This project is a grouped project. The first and second project activity instances, which are the subject of this second verification, include 5,227.25 ha as outlined in section 2.1.

The second project activity instance included in this monitoring period (278.23 ha) meets the eligibility criteria. The eligibility criteria for inclusion of new project activity instances are demonstrated in accordance with paragraph 3.4.9 of the VCS Standard (Version 3.7).

The new project activity instance meets the following criteria:

⁶ Available as supporting documentation as "VCS_Accession-Representation YAGASU 2019"

⁷ See shapefile named "1st_instance_extracted_areas_from_first_verification" included as MR supporting documentation

- 1) Meet the applicability conditions set out in the methodology applied to the project, meaning the methodology AR-ACM0014: Afforestation and reforestation of degraded mangrove habitats, version 03.0:
 - a. The land subject to the project activity is degraded mangrove habitat. The planting areas of this ARR project activity, and the new instance areas, are sited in a coastal location where the population pressure increased the degradation of the mangroves since several decades. For the selection of the project areas, mostly abandoned fishponds were selected as well as degraded coastal and river mudflats. Average crown cover of these areas is lower than 2%. The new instances are within the outer boundary of this grouped project which is subject to the definition of the degraded mangrove habitats.
 - b. More than 90% of the project area is planted with mangrove species. If more than 10% of the project area is planted with non-mangrove species, then the project activity does not lead to alteration of hydrology of the project area and hydrology of connected up-gradient and down-gradient wetland area. Only mangrove species are selected and will be selected for all future project instances. The new instances added in this period have been planted with mangrove species.
 - c. Soil disturbance attributable to the ARR project activity does not cover more than 10% of area. As for the first instances, the plantation techniques for this new instance in this project is not involve soil disturbance, mangrove seedlings have been planted by direct planting of seeds or planting of nursery-raised seedlings.
- 2) Use technologies or measures similar to the specified in the VCS PD for the first instance and apply them in the same manner. New instance consists of wetlands and abandoned fish ponds restoration through planting and assisting natural regeneration using native species and involving local communities.
- 3) Be subject to the baseline scenario of the VCS PD for the specified project activity and geographic area, and therefore, be established on lands where baseline scenario is degraded wetlands. The whole geographic project area within which the project instances shall be located is subject to the baseline scenario determined in the VCS PD. All project instances (first and second) as well as future instances are classified as fish ponds and partially river mudflats.
- 4) Have characteristics with respect to additionality that are consistent with that of the first instance, facing similar barriers. The baseline scenario and the demonstration of additionality are determined for the entirety of the geographic project area within which project activity instances are developed. The second instance presents similar barriers than the first instance project, namely investment barriers, other than economic/financial barriers; technological barriers; and barriers due to social condition. The whole project occurs within the outer boundary delimited, and only fish ponds or very degraded zones are selected for the planting activities. Therefore, all legal characteristics and protections of the areas will be equal.
- 5) Have the project boundary consistent with the physical/geographical boundary of the grouped project. The second instance is developed and implemented within the geographic delineation of the Yagasu outer project boundary. The new instance polygons are georeferenced and compiled in a GIS database system, available as supporting documentation.
- 6) Have documentary evidence of the starting date.
- 7) Be included in the monitoring report with sufficient technical, financial, geographic and other relevant information to demonstrate compliance with the applicable set of eligibility criteria and enable sampling by the VVB.

Also, to demonstrate the eligibility of the second project activity instance under the AFOLU requirements for ARR a GIS study was conducted to show that the project area was not covered by any forest at the time of the planting and was not cleared of native ecosystems within the 10 year period prior the project start date⁸. The methodology is based on analysis of the evolution of the areas according the satellite images from Google Earth. After this analysis the selected polygons to be included in this grouped project sum up to 278.23 ha.⁹

2.4. Safeguards

2.4.1. No Net Harm

This ARR project activity does not require any form of Environmental Impact Assessment (EIA). However, environmental impacts have been analyzed as part of various studies conducted by Yagasu (2009, 2014 and 2019)¹⁰. These studies are available as supporting documentation. No negative impacts have been identified by the project activity.

2.4.2. Local Stakeholder Consultation

During the period 2011 to 2019, a total of 360 meetings have taken place with local communities, institutions (private sectors, universities, NGOs) and governments, attended by 18,000 people. Yagasu team members have been working in their respective areas for many years and the organization's approaches and practices are well known to local communities. Early consultations provided important inputs and laid the foundation for continuing consultations during project implementation. Community and local government demand for the services provided by this project were confirmed through meetings during the preparation missions.

The project assists the communities in obtaining official recognition of such agreements through the local and provincial government agencies. This consultation process and the resulting agreements are intended to increase involvement by local communities. Regulations agreed to between local communities and relevant government agencies, and developed with the project's appropriate technical inputs, give legal and enforceable status to the long-term restoration and community development program. It is likely that the coastal green belt protection strategy introduced through the project will increase livelihood opportunities for local communities, by institutionalizing communal or individual land ownership. Should any issue arise, the problem is handled through the institutionalized Local Stakeholders Forum.

Yagasu has regular and continuous communication with heads of communities and the rest of stakeholders to address possible suggestions and complaints, training activities, dissemination of monitoring reports, etc.

⁸ See section 3.1.6 of AFOLU requirements: https://verra.org/wp-content/uploads/2018/03/AFOLU_Requirements_v3.6.pdf

⁹ See supporting documentation of MR/eligibility analysis

¹⁰ See supporting documentation.

At the same time, Yagasu has regular and continuous communication with Livelihoods Fund.

Participatory processes have been undertaken directly with villagers of local communities involved in the project activities through community group meetings (466 sessions), village meetings (334 sessions) and district meetings (360 sessions), providing adequate information.

3. DATA AND PARAMETERS

3.1. Data and Parameters Available at Validation

Data and parameters available at validation of the grouped project are described in the project description document. The following list includes only data and parameters used in the second verification of the first and second project activity instances.

Table 3 Data and parameters available at validation

Data / Parameter	$\Delta C_{BSL,t}$
Data unit	t CO ₂ -e
Description	Baseline net GHG removals by sinks in year t
Source of data	Calculated
Value applied:	0
Justification of choice of data or description of measurement methods and procedures applied	The project area is stratified into zero baseline and dwarf mangrove baseline stratum considered shrubs according to the AR CDM tool “Estimation of carbon stocks and change in carbon stocks of trees and shrubs in A/R CDM project activities”. Based on this, no removals by sinks are assumed
Purpose of Data	Calculation of ex-ante and ex-post baseline emissions
Comments	-

Data / Parameter	CF_{TREE}
Data unit	t C (t d.m.) ⁻¹
Description	Carbon fraction of tree biomass
Source of data	Default value of AR CDM tool “Estimation of carbon stocks and change in carbon stocks of trees and shrubs in A/R CDM project

	activities" is applied.
Value applied:	0.47
Justification of choice of data or description of measurement methods and procedures applied	According to AR-TOOL14, the above default value (0.47) should be used unless transparent and verifiable information can be provided to justify a different value.
Purpose of Data	Determination of project emission/removals
Comments	-

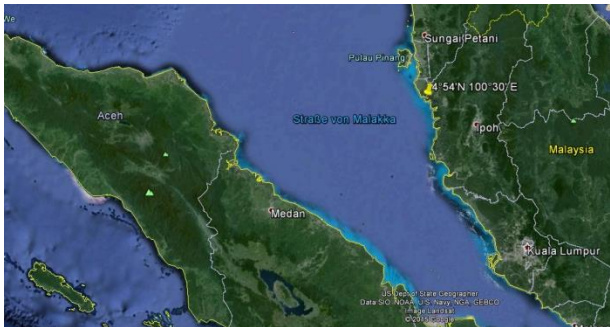
Data / Parameter	R_j
Data unit	Dimensionless
Description	Root-shoot ratio for tree species j
Source of data	Application of the following equation from the AR-TOOL14: $R_j = \exp[-1.085 + 0.9256 \cdot \ln(AGB)] / AGB$ Where AGB is the above-ground tree biomass per hectare (in t d.m. ha ⁻¹)
Value applied:	See database
Justification of choice of data or description of measurement methods and procedures applied	According to AR-TOOL14, root-shoot ratio for tree species shall be calculated with the formula: $R_j = \exp[-1.085 + 0.9256 \cdot \ln(AGB)] / AGB$
Purpose of Data	Calculation of project emission removals
Comments	-

Data / Parameter	$f_j (X_{1,i}, X_{2,i}, X_{3,i}, \dots)$
------------------	--

Data unit	t d.m (tone of dry matter).
Description	Above-ground biomass of the tree returned by the allometric equation for species <i>j</i> relating the measurements of tree <i>i</i> to the above-ground biomass of the tree.
Source of data	For ex-post: Two equations are used for this verification: 1. Specific equation for <i>Rhizophora</i> spp. from Ong et al. (2004) ¹¹ 2. Common allometric equation for other mangroves from <i>Komiyama et al., 2005</i> ¹² .
Value applied:	For ex-post: Rhizophora spp.: <i>Rhizophora apiculata</i> $W_{top} = 0.235DBH^{2.42} \quad r^2 = 0.98, n = 57, D_{max} = 28 \text{ cm}, \text{ Ong et al. (2004)}$ Other mangrove species: Common equation $W_{top} = 0.251\rho D^{2.46} \quad r^2 = 0.98, n = 104, D_{max} = 49 \text{ cm}, \text{ Komiyama et al. (2005)}$ Where: W_{TOP} = Above-ground biomass; kg. D = Diameter (at breast height); cm ρ = wood density; g/cm ³
Justification of choice of data or description of	Ex-post allometric equations are appropriate as determined by the A/R Methodological Tool "Demonstrating appropriateness of

¹¹<http://www.sciencedirect.com/science/article/pii/S0378112703003906> (also provided as supporting documentation)

¹²Komiyama, A.; Pongparn, S. & Kato, S. (2005) Common Allometric equations for estimation the tree weight of mangroves. *Journal of tropical Ecology* 21: 471-477. Cambridge University Press (available as supporting documentation), also in http://www.cifor.org/publications/pdf_files/WPapers/MP86CIFOR.pdf

measurement methods and procedures applied	allometric equations for estimation of aboveground tree biomass in A/R CDM project activities". Since the majority of trees in the project are <i>R. apiculata</i>, this species-specific equation is considered. It is also applied to the other <i>Rhizophora</i> species (<i>Stylosa</i> and <i>Mucronata</i>), which are all stilt mangroves. This equation represents the best option for this first verification since it was derived (Ong et al. 2004) in Malaysia, just opposite the project area in Sumatra (see location on the map below).  Further, it is one of few equations which also included stilt roots in the aboveground biomass. Finally, it is the only equation found which also includes smaller diameters (the study ranged from 1.1 – 28 cm). The common equation developed by Komiyama excludes stilt roots of mangroves and represents one of the most widely used equations. See supporting documentation.
Purpose of Data	Calculation of project emission removals
Comments	-

Data / Parameter	$dSOC_t$
Data unit	$t\ C\ ha^{-1}\ yr^{-1}$
Description	The rate of change in SOC stocks within the project boundary, in year t .
Source of data	
Value applied:	3.32

Justification of choice of data or description of measurement methods and procedures applied	Value calculated based on publication Murdiyarso et al 2015 ¹³ .
Purpose of Data	Calculation of project emission removals
Comments	-

Data / Parameter	EF _{CH4,i}
Data unit	g CH ₄ (kg dry matter burnt) ⁻¹
Description	Emission factor for CH ₄ in stratum i
Source of data	CDM A/R Methodological Tool "Estimation of non-CO2 GHG emissions resulting from burning of biomass attributable to an A/R CDM project activity" (Version 04.0.0)
Value applied:	A default value of 6.8 from the above CDM A/R Methodological Tool is used.
Justification of choice of data or description of measurement methods and procedures applied	A default value is used since values are not available from sources numbered a., b., and c. described below. According to the above CDM A/R Methodological Tool, values may be selected from the following sources, in order of preference: a. Regional/national inventories e.g. national forest inventory, national GHG inventory; b. Inventory from neighboring countries with similar conditions; c. Globally available data applicable to the project site or to the region/country where the site is located; d. Default values

¹³ Murdiyarso, D et al (2015). The potential of Indonesian mangrove forests for global climate change mitigation. Nature climate change. Available as supporting documentation

Purpose of Data	Calculation of ex-post project emissions
Comments	-

Data / Parameter	$EF_{N_2O,i}$
Data unit	$g\ N_2O/(kg\ dry\ matter\ burnt)^{-1}$
Description	Emission factor for N_2O in stratum i
Source of data	CDM A/R Methodological Tool "Estimation of non-CO ₂ GHG emissions resulting from burning of biomass attributable to an A/R CDM project activity" (Version 04.0.0)
Value applied:	A default value of 0.2 from the above CDM A/R Methodological Tool is used.
Justification of choice of data or description of measurement methods and procedures applied	A default value is used since values are not available from sources numbered a., b., and c. described below. According to the above CDM A/R Methodological Tool, values may be selected from the following sources, in order of preference: a. Regional/national inventories e.g. national forest inventory, national GHG inventory; b. Inventory from neighboring countries with similar conditions; c. Globally available data applicable to the project site or to the region/country where the site is located; d. Default values;
Purpose of Data	Calculation of ex-post project emissions
Comments	-

Data / Parameter	GWP_{CH_4}
Data unit	Dimensionless

Description	Global warming potential for CH ₄
Source of data	CDM A/R Methodological Tool "Estimation of non-CO2 GHG emissions resulting from burning of biomass attributable to an A/R CDM project activity" (Version 04.0.0)
Value applied:	21
Justification of choice of data or description of measurement methods and procedures applied	Default value
Purpose of Data	Calculation of ex-post project emissions
Comments	-

Data / Parameter	GWP _{N2O}
Data unit	Dimensionless
Description	Global warming potential for N ₂ O
Source of data	CDM A/R Methodological Tool "Estimation of non-CO2 GHG emissions resulting from burning of biomass attributable to an A/R CDM project activity" (Version 04.0.0)
Value applied:	310
Justification of choice of data or description of measurement methods and procedures applied	Default value
Purpose of Data	Calculation of ex-post project emissions
Comments	-

Data / Parameter	COMF _i										
Data unit	Dimensionless										
Description	Combustion factor for stratum i										
Source of data	CDM A/R Methodological Tool "Estimation of non-CO2 GHG emissions resulting from burning of biomass attributable to an A/R CDM project activity" (Version 04.0.0)										
Value applied:	The corresponding default according to the age is used, unless transparent and verifiable information can be provided to justify a different value: <table border="1"> <thead> <tr> <th>Mean age (years)</th><th>Default value</th></tr> </thead> <tbody> <tr> <td>3-5</td><td>0.46</td></tr> <tr> <td>6-10</td><td>0.67</td></tr> <tr> <td>11-17</td><td>0.50</td></tr> <tr> <td>18 and above</td><td>0.32</td></tr> </tbody> </table>	Mean age (years)	Default value	3-5	0.46	6-10	0.67	11-17	0.50	18 and above	0.32
Mean age (years)	Default value										
3-5	0.46										
6-10	0.67										
11-17	0.50										
18 and above	0.32										
Justification of choice of data or description of measurement methods and procedures applied	A default value is used since values are not available from sources numbered a., b., and c. described below. Default emission factor for tropical forest from the CDM A/R tool. According to the above CDM A/R Methodological Tool, values may be selected from the following sources, in order of preference: a. Project-specific calculation, regional/national inventories e.g. national forest inventory, national GHG inventory; b. Inventory from neighboring countries with similar conditions; c. Globally available data applicable to the project site or to the region/country where the site is located d. Default values										
Purpose of Data	Calculation of ex-post project emissions										
Comments	-										

Data / Parameter	E
Data unit	t d.m. (or t d.m. ha ⁻¹)
Description	Acceptable margin of error (i.e. one-half the confidence interval) in estimation of biomass stock within the project boundary
Source of data	AR-TOOL14
Value applied:	10% of the mean value of biomass stock
Justification of choice of data or description of measurement methods and procedures applied	N/A
Purpose of Data	Calculation of ex-post project emissions
Comments	-

Data / Parameter	t _{val}
Data unit	Dimensionless
Description	Two-sided Student's t-value at infinite degrees of freedom for the required confidence level
Source of data	AR-TOOL14
Value applied:	According to the student's t-distribution table, 1.654 for confidence level 90% and infinite degrees of freedom
Justification of choice of data or description of measurement methods and procedures applied	N/A
Purpose of Data	Calculation of ex-post project emissions
Comments	The value used is for the 90% confidence level for determination of biomass stock prescribed in the A/R CDM project activities since a different confidence level is not prescribed in the methodology.

Data / Parameter	DF _{DW}
Data unit	Percent
Description	Conservative default factor expressing carbon stock in dead wood as a percentage of carbon stock in tree biomass
Source of data	A/R Methodological tool "Estimation of carbon stocks and change in carbon stocks in dead wood and litter in A/R CDM project activities
Value applied:	6%
Justification of choice of data or description of measurement methods and procedures applied	Default value depending on the biome, elevation and precipitation of the area. The tool recommended this value unless transparent and verifiable information can be provided to justify a different value.
Purpose of Data	Calculation of ex-post project emissions
Comments	-

3.2. Data and Parameters Monitored

Table 4 Data and parameters monitored

Data / Parameter	A_i			
Data unit	Ha			
Description	Area of tree biomass stratum i			
Source of data	GIS and GPS			
Description of measurement methods and procedures to be applied	Once the criterion for ex-post stratification has been stratified and A_i has been calculated using GIS.			
Frequency of monitoring/recording	At least for the first monitoring but can be updated in every verification.			
Value monitored	Stratum ID	Description	A_i (ha) First project activity instances	A_i (ha) second project activity 2nd instances
	1	Stratum 1, dense: highest crown cover (>60%) and lower mortality. Planting lines are not visible in the high-resolution image available from Google Earth since the trees have reached the tangency of canopies. This stratum is probably related to the oldest plantations from 2011-2013.	610.09	151.82
	2	Stratum 2, medium: medium crown cover (10-60%) and lower mortality. Planting lines are visible in the high-resolution image available from Google Earth. This stratum is related to plantations from 2013-2015.	2,443.88	110.63
	3	Stratum 3, scattered: low crown cover and high mortality. No plants or poor growth where it is not possible to differentiate the planting lines in the high-	1,895.05	15.77

		resolution image available from Google Earth. This stratum is related to plantations from 2015 or after ¹⁴ .		
	TOTAL		4,949.02	278.23
Monitoring equipment	GIS and other spatial information software			
QA/QC procedures to be applied	N/A			
Purpose of the data	Calculation of project emissions/removals			
Calculation method	Use of GIS tool			
Comments	-			

Data / Parameter	n_i
Data unit	Dimensionless
Description	Number of sample plots in stratum i
Source of data	Calculated
Description of measurement methods and procedures to be applied	N/A
Frequency of monitoring/recording	N_i is calculated for each monitoring event, at least every five years.
Value applied:	The following values are estimated from the first and second project activity instance:

¹⁴ Pending new plantation plot called recalled plots

	Stratum	n_i
	1	75
	2	96
	3	66
		237
Monitoring equipment	N/A	
QA/QC procedures to be applied	N/A	
Purpose of data	Calculation of project emissions/removals	
Calculation method	The calculation method is described in the tool "Calculation of the number of sample plots for measurements within A/R CDM project activities" (version 02.1.0) ¹⁵	
Comments	-	

Data / Parameter	w_i
Data unit	Dimensionless
Description	Relative weight of the area of stratum i , the area of stratum i divided by the project area.
Source of data	Calculated as the area of stratum i divided by the project area.
Description of measurement methods and procedures to be applied	N/A
Frequency of monitoring/recording	Calculated for each monitoring event, at least every five years
Value applied:	The following values are estimated from the first and second project activity instances:

¹⁵<https://cdm.unfccc.int/methodologies/ARmethodologies/tools/ar-am-tool-03-v2.1.0.pdf>

	Stratum	W_i
	1	0.14
	2	0.48
	3	0.37
Monitoring equipment	N/A	
QA/QC procedures to be applied	N/A	
Purpose of data	Calculation of project emissions/removals	
Calculation method	Area of stratum i divided by the project area	
Comments	-	

Data / Parameter	S_i								
Data unit	t d.m. (or td.m. ha ⁻¹)								
Description	Estimated standard deviation of biomass stock in stratum i								
Source of data	Inventory or default value								
Description of measurement methods and procedures to be applied	N/A								
Frequency of monitoring/recording	S_i is calculated for each monitoring event, at least every five years								
Value applied:	The following values are estimated from the first and second project activity instance: <table> <tr> <th>Stratum</th><th>S_i</th></tr> <tr> <td>1</td><td>35.77</td></tr> <tr> <td>2</td><td>14.75</td></tr> <tr> <td>3</td><td>19.34</td></tr> </table>	Stratum	S_i	1	35.77	2	14.75	3	19.34
Stratum	S_i								
1	35.77								
2	14.75								
3	19.34								
Monitoring equipment	N/A								

QA/QC procedures to be applied	N/A
Purpose of data	Calculation of project emissions/removals
Calculation method	Excel or tool available to calculate standard deviation
Comments	-

Data / Parameter	$A_{PLOT,i}$
Data unit	Ha
Description	Size of sample plot in stratum i
Source of data	Calculated
Description of measurement methods and procedures to be applied	N/A
Frequency of monitoring/recording	Every three years since the year of the initial verification
Value applied:	0.009 (5m radius)
Monitoring equipment	N/A
QA/QC procedures to be applied	As this value is calculated based on the expected planting density, during the sampling process it needs to be confirmed that the mean number of trees per sample plot is more than 15.
Purpose of data	Calculation of project emissions/removals
Calculation method	$A_{PLOT,i}$ will be calculated depending on the expected density (trees/ha) in each stratum, with the objective of having around 15 trees per sample plot.
Comments	-

Data / Parameter	$A_{BURN,i,t}$
Data unit	Ha

Description	Area burnt in stratum i
Source of data	Field measurement, remote sensing measurement or any other spatial information available
Description of measurement methods and procedures to be applied	The area is delineated on the ground using GPS or georeferenced remote sensing data or from any other spatial information available.
Frequency of monitoring/recording	The burnt area is measured whenever a forest fire has occurred.
Value applied:	N/A, wildfires did not occur in any mangrove forests in the areas in this monitoring period
Monitoring equipment	GPS
QA/QC procedures to be applied	Field teams are trained in the correct use of GPS. They are also made fully aware of all procedures and the importance of collecting data as accurately as possible.
Purpose of data	Calculation of project emissions/removals
Calculation method	N/A
Comments	Applicable only if wildfires occur.

Data / Parameter	X_i
Data unit	Variable
Description	Variables measured per tree for the calculation of above-ground biomass applying an allometric equation: DBH, height, D_{30} ,
Source of data	Measured
Description of measurement methods and procedures to be applied	<i>DBH and D_{30}</i> readings are taken from a caliper/ diameter tape at 1.3 m along the stem (for DBH) or at 30 cm from the ground for D_{30} . Height – readings are taken on a calibrated pole placed along the longitudinal axis of the tree.
Frequency of monitoring/recording	Measured every monitoring event, at least every five years

Value applied:	n.a.
Monitoring equipment	<i>DBH and D₃₀</i> , a caliper/ diameter tape is used. Height, a calibrated pole is used.
QA/QC procedures to be applied	Field teams are trained in all inventory procedures including correct measurement – as described in the Pilot Inventory, Sampling & Monitoring Plan. Field-team members are fully aware of all procedures and the importance of collecting data as accurately as possible. Finally, a quality check is undertaken as part of inventory to identify and correct errors if any.
Purpose of data	Calculation of project emissions/removals
Calculation method	n.a.
Comments	D ₃₀ , only for the first verification

Data / Parameter	T
Data unit	Year
Description	Time period elapsed between two successive estimations of carbon stock in a carbon pool
Source of data	Recorded time
Description of measurement methods and procedures to be applied	N/A
Frequency of monitoring/recording	N/A
Value applied:	N/A
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Purpose of data	Calculation of project emission removals
Calculation method	N/A

Comments	If the two successive estimations of carbon stock in a carbon pool are carried out at different points of time in year t_2 and t_1 , (e.g. in the month of April in year t_1 and in the month of September in year t_2), then a fractional value will be assigned to T
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3.3. Monitoring Plan

i. Organizational structure

Institutionally, a permanent YAGASU Carbon Survey Team has been set up consisting of several field teams consisting of 2-3 field officers and under the supervision of Restoration Manager (Mr. Anton Siregar) and Research and Field Monitoring Manager (Ms. Grace Yanti Panjaitan). These teams will undertake all surveys in the project to ensure consistency in measurements and will implement the carbon monitoring inventory of permanent sampling points (PSPs). The field measurement is supported by a project-based Android data collection system (YAGASU survey App) which ensures standardized data collection with QA/QC functions to minimize errors during this stage.

Further, YAGASU and Livelihoods have set up a stringent verification system with external tree audits including annual boundary verification and revision, if required.

The organizational structure of the monitoring is divided into two layers. The first layer is represented by the YAGASU field staff, who are trained in all necessary activities to perform the forest inventory, boundary tracking with GPS, socio-economic monitoring and forest establishment monitoring (survival rate, nursery monitoring, etc.). More than 20 field officers from YAGASU are responsible for the different districts of the project areas ranging from east to west of the northern coast of Sumatra. Generally, they are part of the communities and are well acquainted with the specific conditions within the different planting areas. The forest inventory and monitoring surveys are conducted by them in the field (as part of the carbon survey teams). All results are directly brought to the main office of YAGASU in Medan: the second layer of the project. Bambang Suprayogi and the YAGASU technical program coordinators are responsible for the technical implementation of the whole project. In the main office the data are processed, analyzed and archived following standard operation procedures and good practice guidelines. To guarantee a high level of certainty of the results, the YAGASU technical program coordinators will periodically crosscheck the data in the field as an independent survey. It is envisaged to verify at least 10% of the data after each inventory or survey is conducted.

The project implementation is based on the local presence of YAGASU staff in the project area. The main role of the field officers is to manage the reforestation/restoration activity in close cooperation with YAGASU technical program coordinators.

- Selecting randomly and verifying the GPS location of at least 10% of the plots planted during a particular planting season.
- Comparing the trees planted with the trees recorded in the planting plans
- Assessing the degree (in %) of survival of the mangrove seedlings and preparing a report with the findings considering a minimum precision of 10% at the 90% confidence level.

Replanting of mangroves is only necessary if the optimal tree density of 3,000 trees per ha cannot be achieved due to very high natural mortality. This has to be decided on a site-by-site basis due to the varying local tidal and ecological conditions.

- Area verification. Project parcels will be verified using GPS in the field as well as through Google Earth imagery analysis.

The Livelihoods and Yagasu's mangrove restoration grouped project in The East Coast of Aceh and North Sumatra Province is part of the portfolio of the Livelihoods Fund and for the monitoring process it has followed the General Monitoring for Project Developers¹⁶, the Livelihoods Standard Operating Procedures. These procedures consist of several guidance documents for different monitoring phases:

- Mapping and stratifying
- Inventory design
- Field work planning
- Field work

This document complements those general procedures with the description of the specific characteristics of the monitoring process of the Sumatra project, based on the above mentioned guidance and adapted to the project, and with specific procedures for data recording and quality assessment & quality checks.

ii. Stratifying

Summary

For this second verification, new project stratification has been performed. In the first monitoring, the project area was stratified in four strata based on growth and survival rate. However, this stratification is not consistent nowadays with the state of the vegetation during the second verification, as has been checked using high-resolution Google Earth images. Furthermore, in the first stratification each planted parcel was considered in one project strata and it has been demonstrated that there is a gradient of ecological conditions (mainly salinity) inside every planted parcel depending on the distance to the closest water stream. This issue has led to a high heterogeneity inside each plantation polygon having as a consequence a high sampling uncertainty. For these reasons and in order to reduce the sampling error, it has been decided to carry out a new stratification based on a supervised classification of Sentinel-2 satellite images.

Remote sensing data

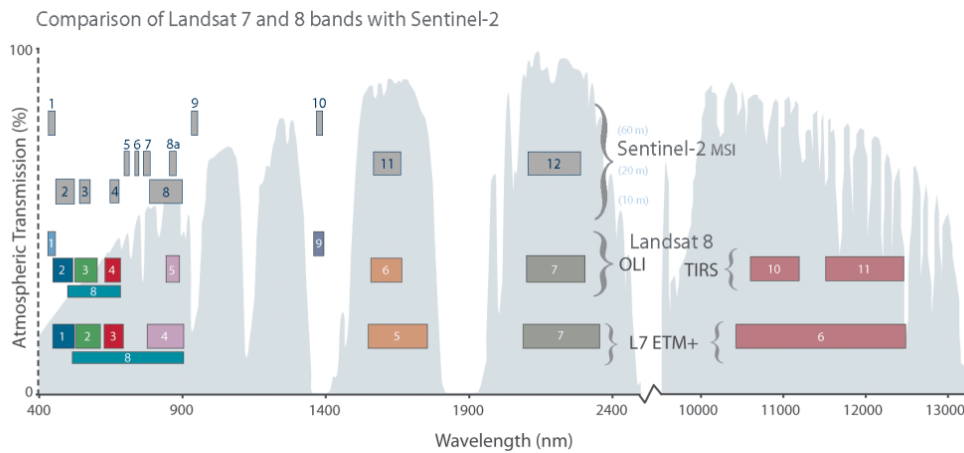
The different soil and salinity conditions in the project area mean that the spatial variability of aboveground biomass is high even within the same plantation plot. In order to improve the forest inventory estimates, stratification has been carried out based on the automatic classification of Sentinel-2A satellite images.

The Sentinel-2 mission of the European Space Agency (ESA) consists of two twin satellites: Sentinel 2A, launched on 23 June 2015 and Sentinel-2B, launched on 7 March 2017. The launch

¹⁶ Available as supporting documentation as "2013.09.06 General Monitoring for Project Developers"

of these satellites is part of ESA's Copernicus Programme. Sentinel-2 represents a high-resolution multispectral mission designed to monitor the Earth's surface by analyzing the vegetation cover. The temporal resolution of each of the satellites is 10 days, 5 days considering both satellites. It consists of 13 spectral bands, four bands at 10 m, six bands at 20 m and three bands at 60 m.

Figure 3 Identification of Sentinel-2 bands with respect to Landsat 7 and 8 bands. (Source: <http://landsat.gsfc.nasa.gov/>)



As can be seen in the figure the Sentinel-2 bands are like the Landsat 7 and 8 bands, except for the thermal band of the spectrum. In addition to extending the spectral and temporal resolution, Sentinel-2 offers bands at different spatial resolutions of 10, 20 and 60 m. The size of each Sentinel-2 image is 100 x 100 Km².

Sentinel-2 images with a processing level of 1C have been used for the creation of the free cloud mosaic. This level of processing provides geometrically corrected images with reflectance values on the roof of the atmosphere.

Image classification

Random forest (Breiman 2001, henceforth RF¹⁷) classification models have been applied in the stratification process. RF is part of the automatic learning methods and has become a widespread technique in vegetation mapping and prediction of forest variables.

RF is a machine learning technique based on the ensemble of many decision trees. For each individual decision tree, data is randomly segregated into two data sets for training and validation. The procedure used in the random generation of decisions allows low correlation between the individual decision trees, ensuring robustness in the RF results. Several advantages have been observed comparing RF with other machine learning alternatives, such as the capacity for

¹⁷ Breiman, L., 2001. Random Forests. Mach. Learn. 45, 5–32. <https://doi.org/10.1023/A:1010933404324>

working with numerous predictor variables, the lack of overfitting problems, and robustness with respect to noise in the data.

The training sample has been generated using high-resolution images available in Google Earth.

Sentinel-2 spectral bands and vegetation indices derived from these bands have been used as predictive covariables in the classification. As can be seen in Figure 1, Sentinel-2 adds information on the wavelengths between red and infrared, an area of the electromagnetic spectrum where vegetation has a very different spectral behavior with respect to other land cover. The following table shows the predictor variables used in the classification.

Table 5 Variables used for stratification

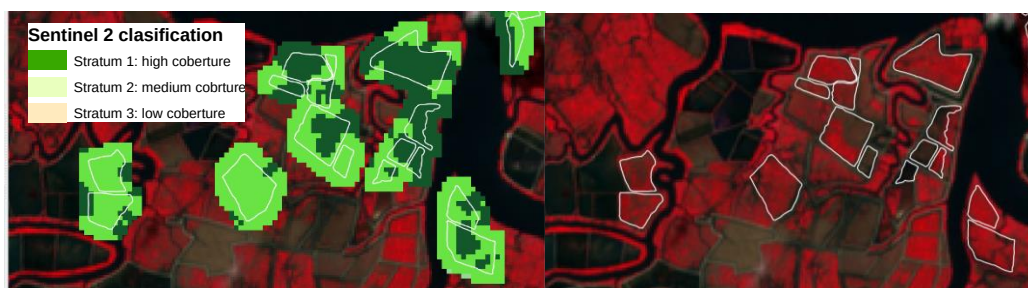
Variable types	Number of covariable	Covariable	Sentinel-2 bands
Spectrals	1	Blue	B2
	2	Green	B3
	3	Red	B4
	4	Red-edge	B5
	5	Red-edge	B6
	6	Red-edge	B7
	7	Near infrared (NIR)	B8
	8	Near infrared (NIR)	B8A
	9	Near infrared (SWIR-1)	B11
	10	Near infrared (SWIR-2)	B12

Variable types	Number of covariable	Covariable	Definition	Sentinel-2 bands
Vegetation index	11	NDVI1 (Rouse et al. (1973))	$(NIR - R)/(NIR + R)$	$(B7 - B4)/(B7 + B4)$
	12	NDVI2	$(NIR - R)/(NIR + R)$	$(B8 - B4)/(B8 + B4)$

Variable types	Number of covariable	Covariable	Definition	Sentinel-2 bands
		(Rouse et al. (1973))		
	13	NDI45 (Delegido et al. (2011	$(\text{NIR} - \text{R})/(\text{NIR} + \text{R})$	$(\text{B5}-\text{B4})/(\text{B5} + \text{B4})$
	14	SAVI	$(1+\text{L}) * (\text{NIR} - \text{R})/(\text{NIR} + \text{R} + \text{L})$	$(1+\text{L}) * (\text{B7}-\text{B4})/(\text{B7} + \text{B4} + \text{L})$
	15	TCARI (Haboudane et al. (2002))	$3 * [(\text{RE}-\text{R}) - 0.2 * (\text{RE}-\text{G}) * (\text{RE}/\text{R})]$	$3 * [(\text{B5}-\text{B4}) - 0.2 * (\text{B5}-\text{B3}) * (\text{B5}/\text{B4})]$
	16	OSAVI	$(1.16) * (\text{NIR}-\text{R})/(\text{NIR} + \text{R} + 0.16)$	$(1.16) * (\text{B7}-\text{B4})/(\text{B7} + \text{B4} + 0.16)$
	17	MTCI (Dash and Curran (2004))	$(\text{NIR}-\text{RE})/(\text{RE}-\text{R})$	$(\text{B6}-\text{B5})/(\text{B5}-\text{B4})$
	18	MCARI (Daughtry et al. (2000))	$[(\text{RE}-\text{R}) - 0.2 * (\text{RE}-\text{G})] * (\text{RE}-\text{R})$	$[(\text{B5}-\text{B4}) - 0.2 * (\text{B5}-\text{B3})] * (\text{B5}-\text{B4})$
	19	GNDVI (Gitelson et al. (1996))	$(\text{NIR}-\text{G})/(\text{NIR} + \text{G})$	$(\text{B7}-\text{B3})/(\text{B7} + \text{B3})$
	20	PSSRa (Blackburn (1998))	NIR/R	$\text{B7}/\text{B4}$
	21	S2REP (Frampton et al, 2013)	$705 + 35 * (((\text{NIR} + \text{R})/2 - \text{RE1})/(\text{RE2} - \text{RE1}))$	$705 + 35 * (((\text{B7} + \text{B4})/2 - \text{B5})/(\text{B6} - \text{B5}))$
	22	IRECI (Frampton et al, 2013)	$(\text{NIR}-\text{R})/(\text{RE1}/\text{RE2})$	$(\text{B7}-\text{B4})/(\text{B5}/\text{B6})$

The "salt and pepper" effect of the classification has been refined by using a majority filter with a 3x3 window. The result has been converted to shape format in order to facilitate further work. Figure 2 shows a detail of the final classification.

Figure 4 Example of the classification on the Sentinel-2 image



Proposed stratification

The three strata considered are listed below:

- Stratum 1, dense: highest crown cover (>60%) and lower mortality. Planting lines are not visible in the available Google Earth high-resolution image since the trees have reached the tangency of canopies. This stratum is probably related to the oldest plantations from 2011-2013.
- Stratum 2, medium: medium crown cover (10-60%) and lower mortality. Planting lines are visible in the available Google Earth high-resolution image. This stratum is related to plantations from 2013-2015.
- Stratum 3, scattered: low crown cover and high mortality. No plants or poor growth where it is not possible to differentiate the planting lines in the available Google Earth high-resolution image. This stratum is related to plantations from 2015 or after 2011.

Table 6 Stratification of first and second project activity instances.

	Area (ha) in first project activity instance	Area (ha) in second project activity instance
Stratum 1	610.09	151.82
Stratum 2	2443.88	110.63
Stratum 3	1,895.05	15.77
Total	4,949.02	278.23

The second project activity instance includes plantation areas which were planted between 2011 and 2017 but they were not included in the first verification period due to economic reasons. A summary of planted areas in both first and second project activity instances is shown below:

Table 7 Summary of planting area for the first and second project activity instances.

Planting year	First project activity instance	Second project activity instance
	Area (ha)	Area (ha)
2011	191.18	7.74
2012	851.32	51.95
2013	1,986.53	12.73
2014	1,623.19	169.90
2015	296.80	1.34
2016		26.35
2017		8.21
Total	4,949.02	278.23

iii. Inventory design

Sampling design

Due to the heterogeneity of biomass carbon stocks in the project area, the total number of necessary sample plots to get a sampling error lower than 10% is too high. As the measurement of a high number of sample plots was not feasible the project proponent decided to estimate biomass carbon stocks with a higher uncertainty than 10% and apply an uncertainty discount to the monitoring results following Appendix 2 of AR-TOOL14.

A minimum number of 40 or 60 plots per stratum should be established. Based on the expert criteria and previous experience, a systematic sample was distributed on a grid of 300 m by 300 m in Stratum 1 and 500 by 500 m in Stratum 2 and 3. This resulted in a total of 242 sample plots: 77 in stratum 1; 96 in stratum 2 and 69 in stratum 3.

Table 8 Number of sample plots per stratum used in the monitoring

Stratum ID	Description	A _i (ha)	n _i
1	Low crown cover and high mortality (< 10%)	761.92	74
2	Higher crown cover (10-60%) and lower mortality	2,554.51	95
3	Highest crown cover (60-100%)	1,910.82	66
Total		5,227.25	237

Data collection

A detailed description of the YAGASU carbon inventory procedures can be found in the SOPs (supporting documentation) including standard operating procedures on work safety, field measurement planning and organization, navigating in the field, and tree measurement procedures, etc.

In the field, several parameters are assessed/ measured. The parameters relevant for the carbon monitoring are the following:

- Species name
- Origin: Classification of the tree according to the origin of the tree: Pre-existing tree, Planted tree or Natural regeneration.
- Size: Classification of trees and saplings.
- Diameter: Diameter at breast height (1.3 m above root collar) is measured if available, otherwise the diameter D30 is taken, 30 cm above the root collar.
- Total Height: Total tree height of the tree or sapling.

For navigation and data entry of tree and plot variables, a smartphone-based inventory application was specifically designed for this project.

Figure 5 Example of visualization of data collection on the smartphone-based inventory application

The screenshot shows a smartphone screen with the 'Record Tree Data' application. At the top, there's a header with 'PS11P01_810_37' and 'Record Tree Data'. Below this are three buttons: a green '+ Save' button, a blue 'COMPASS' button, and a black 'End' button. The form fields are as follows:

- Size of Plant:** Two buttons, 'Tree' (selected) and 'Seedling'.
- Origin:** Three buttons, 'Project' (selected), 'Natural', and 'Pre-Exist'.
- Species:** Four buttons, 'Apiculata', 'Mucronata' (selected), 'Stylosa', and 'Avicennia'. There is also an 'Other' button below them.
- Fix Azimuth:** A text input field containing '343'.
- Ground distance (m):** A text input field containing '2'.
- Tree Height (m):** A text input field containing '2'.
- Diameter:** Three buttons, 'DBH' (selected), 'D30cm', and 'D10cm'.
- Add Diameter box:** A text input field containing '1.9'.
- Remove:** A text input field containing '2.4'.
- Diameter deviation (cm):** A text input field containing '0'.

iv. Monitoring of net anthropogenic GHG removals

Ex post estimation of the baseline net GHG removals by sinks

The baseline net GHG removals by sinks are estimated following AR-TOOL14, as indicated in the PD; therefore, they have not been monitored.

Ex post estimation of the actual net GHG removals by sinks

For the monitoring of GHG removals by sinks the project proponent has put in place within Yagasu a unit dedicated to the monitoring processes. This unit benefits from the expertise of a forest carbon specialist who is in charge of the design and calculations of the monitoring inventory.

The monitoring of the GHG removals has been based on the stratification of the project and the sampling design.

Ex post estimation of the actual net GHG removals by sinks has been done as explained in the following paragraphs.

- Changes in carbon stocks in tree and shrub biomass: once the project area is stratified and the sampling design is accomplished, ex-post changes in carbon stocks in tree and shrub biomass have been estimated according to

AR-TOOL14. Specific allometric equations have been developed for the project. Based on those equations the following variables have been measured in all trees inside each sample plot (see further details in the SOP, available as supporting documentation):

- Ht = Total height; cm
- Dmean = Mean crown cover diameter; cm
 - Changes in carbon stocks in deadwood have not been monitored because they have been calculated using the conservative default-factor based method of the CDM tool AR-TOOL12.
 - Changes in carbon stocks in the soil organic carbon pool have been estimated using the default methods proposed in the selected AR CDM methodology (AR-AM00014), therefore no monitoring has been done of this carbon pool.
 - GHG emissions within the project boundary: the only source of GHG emissions considered in the selected methodology is burning of biomass attributable to the project activity and it is calculated using the A/R CDM Methodological tool: "Estimation of non-CO2 GHG emissions resulting from burning of biomass attributable to an A/R CDM project activity, (Version 04.0.0)". In this grouped project there will not be fire for site preparation or fire to clear the land of harvest residue prior to replanting of the land or other forest management. The only potential source of GHG emissions is forest fires that are monitored during the verification period.

Ex post estimation of leakage

In the first project activity instance displacement of pre-project agricultural activities has not occurred. No specific data needs to be monitored for the assessment of this source of leakage.

Ex post estimation of the net anthropogenic GHG removals

The net anthropogenic greenhouse gas removals by sinks have been calculated based on the selected methodology as the actual net greenhouse gas removals by sinks minus the baseline net greenhouse gas removals by sinks minus leakage as appropriate.

Quality control and quality assurance (QA/QC) procedures

A rigid quality control and quality assurance (QA/QC) method is implemented to ensure: (i) reliability of collection of field measurements, (ii) verification of the methods used to collect field data, and (iii) verification of data maintenance and archiving (see further details in the SOP, available as supporting documentation).

Procedures to ensure reliable field measurements

Collecting reliable field measurement data is an important step to ensure the quality of field measurements and therefore is an important step in the quality assurance plan. In this monitoring, field measurements have been done in some trees for the QC/QA of the field information. Those responsible for the measurement work have been trained in all aspects of the field data collection and data analyses. Standard Operating Procedures (SOPs) have been developed including all

steps of the field measurements, and this document has been shared with field teams. These SOPs describe in detail all steps to be taken for the field measurements (including standard field forms) and contain provisions for documentation for verification purposes so that future field personnel can check past results and repeat the measurements in a consistent way. To ensure the quality of the standard operating procedures, they have been elaborated by third party experts, in close collaboration with the implementing partner. To ensure the collection and maintenance of reliable field data:

- Field-team members are fully aware of all procedures and of the importance of collecting data as accurately as possible; and
- Field teams install test plots if needed in the field and measure all pertinent components using the SOPs to estimate measurement errors;
- The document will list all names of the field team and the project leader will certify that the team is trained;
- New staff adequately trained.

Procedures to verify field data collection

. To verify the quality of data recorded a deep revision of data has been performed. This revision consists on checking every data collected is under logical intervals. For example the following rules have been applied:

Table 9 Example of rule applied in the QC procedures

Field name	Rule applied
tr_azimuth	Maximum azimuth should be 360. If there are 4 trees each tree should be in its correspondent quadrant
tr_hori_distance_m	Each data collected should be under the radius of the plot: 5.6 m.
tr_height_m	Heights over 15 meters should be analysed
tr_dbh_cm	Diameters should have values under 30 cm. higher diameters should be revised

The error in diameter is considered one of the most important. After the first submission of monitored plots a total number of 19 plots were repeated of a total number of 237 (8 % of the plots were repeated). These can be found on the calculations sheets as "redo.1st instance" in the

sheet named as tree¹⁸. Also in the calculation sheets is indicated all the corrections made. over the data collected.”

Further on the VVB go with the field staff and re-measure the 10% of the plots. The following quality targets should be achieved for the re-measurements, compared to the original measurements:

- Missed or extra trees → no error within the plot
- Tree species or groups → no error
- DBH of tree measurements → $< \pm 0.5$ cm or 3 % whichever is greater

Field data collected at this stage will be compared with the original data. Any errors found will be corrected and recorded. Any errors discovered will be expressed as a percentage of all plots that have been re-checked to provide an estimate of the measurement error¹⁹.

Data maintenance and storage

Because of the relatively long-term nature of these project activities, data archiving (maintenance and storage) will be an important component of the work. Data archiving should take several forms and copies of all data should be provided to each project participant.

Copies (electronic) of all field data, data analyses, and models; estimates of the changes in carbon stocks and corresponding calculations and models used; any GIS products; and copies of the measuring and monitoring reports should all be stored in a dedicated and safe place, preferably offsite.

Given the time frame over which the project activity will take place and the pace of production of updated versions of software and new hardware for storing data, it is recommended that the electronic copies of the data and report be updated periodically or converted to a format that could be accessed by any future software application. Copies of all raw data, reports of analysis and supporting spreadsheets will be stored in a dedicated long-term electronic archive for at least 2 years following the end of the last crediting period.

¹⁸ See calculations sheet on supporting documentation

¹⁹ The results are available as supporting documentation, “2019_11_06_YAGASU_Ex_post_Calculations_QC_ sept 2019”. The sheet “Comparative 1” shows the differences between the original plots and the remeasured ones, where it can be seen that the error on dbh is less than 3%.

4. QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1. Baseline Emissions

Under the applicability conditions of the applied methodology AR-AM0014 “Afforestation and reforestation of degraded mangrove habitats” (Version 03.0), it is expected that the baseline carbon stocks in litter and soil organic carbon pools will not show a permanent net increase.

The baseline net GHG removals by sinks are therefore calculated using Equation 1 of the methodology:

$$\Delta C_{BSL,t} = \Delta C_{TREE-BSL,t} + \Delta C_{SHRUB-BSL,t} + \Delta C_{DW-BSL,t} \quad \text{Equation 1 of this MR, 1 of AR-AM0014}$$

Where:

- $\Delta C_{BSL,t}$ = Baseline net GHG removals by sinks in year t; t CO₂-e
- $\Delta C_{TREE_BSL,t}$ = Change in carbon stock in baseline tree biomass within the project boundary in year t; t CO₂-e
- $\Delta C_{SHRUB_BSL,t}$ = Change in carbon stock in baseline shrub biomass within the project boundary in year t; t CO₂-e
- $\Delta C_{DW_BSL,t}$ = Change in carbon stock in baseline dead wood biomass within the project boundary, in year t; t CO₂-e

As indicated in Section 3.1 of the PD, one baseline stratum was defined for the complete project area. The land cover classes were:

- Blank areas: 97.59%
- Dwarf mangrove shrubs: 0.99%
- Pre-existing trees: 1.42%

Shrub baseline biomass

Shrub carbon stock changes in this stratum have been estimated as zero as explained in PD. Baseline carbon stocks in shrubs are calculated following the equation of the AR-TOOL14 “Estimation of carbon stocks and change in carbon stocks of trees and shrubs in A/R CDM project activities” (Version 4.1):

$$C_{SHRUB,T} = \frac{44}{12} * CF_s * (1 + R_s) * \sum_i A_{SHRUB,i} * b_{SHRUB,i} \quad \text{Equation 2 of this MR, 26 of AR-TOOL14}$$

$$b_{SHRUB,t} = BDR_{SF} * b_{forest} * CC_{SHRUB,i} \quad \text{Equation 3 of this MR, 27 of AR-TOOL14}$$

Where:

$C_{SHRUB,t}$ = Carbon stock in shrubs within the project boundary in the baseline; t CO₂e

CF_s = Carbon fraction of shrub biomass, t C (t d.m.)⁻¹

R_s = Root-shoot ratio for shrubs; dimensionless

$A_{SHRUB,i}$ = Area of shrub biomass estimation stratum i, ha

$b_{SHRUB,i}$ = Shrub biomass per hectare in shrub biomass estimation stratum i, t d.m.ha⁻¹

BDR_{SF} = Ratio of shrub biomass per hectare in land having a shrub crown cover of 1.0 and the default above-ground biomass content per hectare in forest in the region where the project is located.

b_{FOREST} = Default above-ground biomass content in forest in the region/country where the A/R CDM project activity is located, t d.m.ha⁻¹

CC_{SHRUB} = Crown cover of shrubs in shrub biomass estimation stratum i at the time of estimation, expressed as a fraction

Crown cover (CC_{SHRUB}) = Following the shrub classification (see PD), 1% of the project areas are covered with shrubs.

b_{FOREST} = For the calculation of this second monitoring period, the default value for Indonesia presented in IPCC table 3.A.1.4. is applied.

b_{FOREST} = 136.0 t d.m. ha⁻¹

For the following parameters, default values taken from the tool AR-TOOL14 are applied:

Table 10 Default values applied

Parameter	Denotation	Value
Carbon fraction of shrub biomass	CF_s	0.47
Root-shoot ratio for shrubs	R_s	0.4
Ratio of shrub biomass per hectare in land having a shrub crown cover of 1.0	BDR_{SF}	0.1

Pre-existing tree baseline biomass

The estimation of carbon stock in pre-project tree biomass was done following equation 20 and 21 of the methodological tool "Estimation of carbon stocks and change in carbon stocks of trees and shrubs in A/R CDM project activities" (Version 04.1):

$$C_{TREE_BSL} = \sum_{i=1} C_{TREE_BSL,i}$$

Equation 4 of
this MR, 20 of
AR-TOOL14

$$C_{TREE_BSL,i} = \frac{44}{12} \times CF_{TREE} \times b_{FOREST} \times (1 + R_{TREE}) \times CC_{TREE_BSL,i} \times A_i$$

Equation 5 of
this MR, 21 of
AR-TOOL14

Where:

C_{TREE_BSL} = Carbon stock in pre-project tree biomass; t CO₂-e

$C_{TREE_BSL,i}$ = Carbon stock in pre-project tree biomass in stratum i; t CO₂-e

CF_{TREE} = Carbon fraction of tree biomass (t C (t.d.m.)⁻¹)

b_{forest} = Mean above-ground biomass in forest in the region or country where the A/R CDM project is located; t d.m.ha⁻¹

R_{TREE} = Root-shoot ratio for trees in the baseline; dimensionless

$CC_{TREE_BSL,i}$ = Crown cover of trees in baseline stratum i, at the start of the A/R CDM project activity, expressed as a fraction; dimensionless

A_i = Area of baseline stratum i, delineated on the basis of tree crown cover at the start of the A/R CDM project activity; ha

Crown cover (CC_{tree})= According to the crown cover analysis, the crown cover of pre-existing trees in baseline represents 1.42% of the area in the project area.

b_{FOREST} =The same value as for the shrubs baseline was applied.

$b_{FOREST} = 136.0 \text{ t d.m. ha}^{-1}$

For the following parameters, default values taken from the tool AR-TOOL14 are applied:

Table 11 Default values applied

Parameter	Denotation	Value
Carbon fraction of tree biomass	CF_T	0.47
Root-shoot ratio for tree	R_T	0.25

Dead wood of pre-existing trees in the baseline

According to the methodological tool AR-TOOL12 “Estimation of carbon stocks and change in carbon stocks in dead wood and litter in AR CDM project activities” (Version 03.1), when there are pre-existent trees in the baseline, dead wood carbon shall be accounted as baseline carbon.

The baseline carbon in dead wood is calculated using the conservative default-factor based method included in AR-TOOL12. The following equation is applied:

$$C_{DW,i,t} = C_{TREE,i,t} * DF_{DW} \quad \text{Equation 6 of this MR, 9 of AR-TOOL12}$$

Where:

- $C_{DW,i,t}$ = Carbon stock in dead wood in stratum i at a given point of time in year t ; t CO₂-e;
- $C_{TREE,i,t}$ = Carbon stock in trees biomass in stratum i at a point of time in year t ; t CO₂-e;
- DF_{DW} = Conservative default factor expressing carbon stock in dead wood as a percentage of carbon stock in tree biomass; per cent.

As stated in the PD, the conservative default factor for dead wood biomass for this project according to the corresponding biome, elevation and annual precipitation is 6% of carbon stock of pre-existing tree biomass. Hence, this value is applied.

The pre-existing baseline carbon estimation for the second project activity instance has been calculated since the second project activity instance has been added in this verification period. The pre-existing baseline carbon stock for the second activity instances is summarized below.

Table 12 Calculation of pre-existing baseline carbon stocks in trees and shrubs. Second project activity instances

Year	Project year	Planted areas/ restored per year (ha)	$C_{TREE,t1}$ Pre-existing biomass of trees (tCO ₂ e)	C_{DW_BSL} Dead wood Pre-existing trees (tCO ₂ e)	C_{SHRUB_BSL} Pre-existing shrub biomass (tCO ₂ e)	C_{BSL} Total biomass baseline (t CO ₂ e)
2011	1	7.74	32.21	1.93	2.50	36.64
2012	2	51.95	216.17	12.97	16.80	245.94
2013	3	12.73	52.98	3.18	4.12	60.27
2014	4	169.90	706.92	42.42	54.93	804.26
2015	5	1.34	5.60	0.34	0.43	6.37
2016	6	26.35	109.64	6.58	8.52	124.74
2017	7	8.21	34.16	2.05	2.65	38.86
Total		278.23	1157.67	69.46	89.95	1317.08

4.2. Project Emissions

The actual net GHG removals by sinks have been calculated using equation 2 of the methodology: AR-AM0014: Afforestation and reforestation of degraded mangrove habitats (Version 03.0), as described below.

$$\Delta C_{ACTUAL,t} = \Delta C_{P,t} - GHG_{E,t} \quad \text{Equation 7 of this MR, 2 of AR-AM0014}$$

Where:

$\Delta C_{ACTUAL,t}$ = Annual actual net greenhouse gas removals by sinks at time t ; t CO₂-e yr⁻¹

$\Delta C_{P,t}$ = Change in carbon stocks in project, occurring in the selected carbon pools, at time t ; t CO₂-e yr⁻¹

$GHG_{E,t}$ = Increase of non-CO₂ GHG emissions within the project boundary as a result of the implementation of the A/R CDM project activity, in year t , t CO₂-e

Change in carbon stocks in project - $\Delta C_{P,t}$

Change in the carbon stocks in the project have been calculated using equation 3 of the methodology:

$$\Delta C_{P,t} = \Delta C_{TREE_PROJ,t} + \Delta C_{SHRUB_PROJ,t} + \Delta C_{DW_PROJ,t} + \Delta SOC_{PROJ,t}$$

Equation 8 of this MR , 3 of AR-AM0014

Where:

$\Delta C_{P,t}$ = Change in carbon stocks in project, occurring in the selected carbon pools, at time t ; t CO₂-e yr⁻¹

$\Delta C_{TREE_PROJ,t}$ = Change in carbon stock in tree biomass in project in year t , as estimated in AR-TOOL14; t CO₂-e yr⁻¹

$\Delta C_{SHRUB_PROJ,t}$ = Change in carbon stock in shrub biomass in project in year t , as estimated in AR-TOOL14; t CO₂-e yr⁻¹

$\Delta C_{DW_PROJ,t}$ = Change in carbon stock in dead wood in project in year t , as estimated in the tool "Estimation of carbon stocks and change in carbon stocks in dead wood and litter in A/R CDM project activities", t CO₂-e

$\Delta SOC_{PROJ,t}$ = Change in carbon stock in the soil organic carbon (SOC) pool within the project boundary, as estimated in AR-AM0014, in year t ; t CO₂-e yr⁻¹

Change in carbon stock in tree biomass - $\Delta C_{TREE_PROJ,t}$

$\Delta C_{TREE_PROJ,t}$, in this grouped project, has been calculated using sections 6, 7 and 8 of AR-TOOL14. Specifically, in this monitoring period, estimations of change in carbon stock in trees between two points of time have been done using the "Difference of two independent stock estimations" method (see section 6.1 of AR-TOOL14). The application of this method has been

based on measurements of sample plots to estimate carbon stock in trees at a point of time (see section 8.1 of AR-TOOL14).

The following equations have been used:

$$\Delta C_{TREEPROJ,t} = \frac{\Delta C_{TREE}}{T}$$

Equation 9 of this MR, 11 of AR-TOOL14

$$\Delta C_{TREE} = C_{TREE,t2} - C_{TREE,t1}$$

Equation 10 of this MR, 1 of AR-TOOL14

Where:

$\Delta C_{TREE_PROJ,t}$ = Change in carbon stock in tree biomass in project in year t ; t CO₂-e yr⁻¹

ΔC_{TREE} = Rate of change in carbon stock in tree biomass within the project boundary during the period between a point of time in year $t1$ and a point of time in year $t2$; t CO₂-e yr⁻¹

T = Time elapsed between two successive estimations ($T=t_2 - t_1$); yr

The next equations are also used:

$$C_{TREE,t} = \frac{44}{12} * CF_{TREE} * B_{TREE,t}$$

Equation 11 of this MR, 12 of AR-TOOL14

$$B_{TREE,t} = b_{TREE,t} * A$$

Equation 12 of this MR, 13 of AR-TOOL14

$$b_{TREE,t} = \sum_{i=1}^M w_i * b_{TREE,t,i}$$

Equation 13 of this MR, 14 of AR-TOOL14

$$b_{TREE,t,i} = \sum_{p=1}^{n_i} b_{TREE,p,t,i} / n_i$$

Equation 14 of this MR, 16 of AR-TOOL14

$$b_{TREE,p,t,i} = B_{TREE,p,t,i} / A_{PLOT,i}$$

Equation 15 of this MR, 1 of Appendix 1, AR-TOOL14

$$B_{TREE,p,t,i} = \sum_j B_{TREE,j,p,t,i}$$

Equation 16 of this MR, 2 of Appendix 1, Ar-TOOL14

$$B_{TREE, j, p, t, i} = \sum_l B_{TREE, l, j, p, t, i}$$

Equation 17 of this MR, 3 of Appendix 1, Ar-TOOL14

$$B_{TREE, l, j, p, t, i} = f_j(X_{1,l}, X_{2,l}, X_{3,l}, \dots) * (1 + R_j)$$

Equation 18 of this MT, 4 of Appendix 1, AR-TOOL14

Where:

$C_{TREE, t}$ = Carbon stock in tree biomass within the project boundary at a point of time in year t ; t CO₂-e. Note: for the first verification, the variable $C_{TREE, t}$ will be set equal to zero based on the results of section 3.1.

CF_{TREE} = Carbon fraction of tree biomass; t C (t d.m.)⁻¹.

A default value of 0.47 is used unless transparent and verifiable information can be provided to justify a different value.

$B_{TREE, t}$ = Total tree biomass within the project boundary at a given point of time in year t ; t d.m.

$b_{TREE, t}$ = Mean tree biomass per hectare within the project boundary at a given point of time in year t ; t d.m. ha⁻¹

A = Project area; ha

w_i = Ratio of the area of stratum i to the sum of areas of tree biomass estimation strata (A_i/A), dimensionless

$b_{TREE, t, i}$ = Mean tree biomass per hectare in stratum i at a given point of time in year t ; t d.m. ha⁻¹

n_i = Number of sample plots in stratum i

$b_{TREE, p, t, i}$ = Tree biomass per hectare in sample plot p of stratum i at a given point of time in year t ; t d.m. ha⁻¹

$B_{TREE, p, t, i}$ = Tree biomass in sample plot p of stratum i at a given point of time in year t ; t d.m.

$A_{PLOT, i}$ = Size of sample plot in stratum i ; ha

$B_{TREE, j, p, t, i}$ = Biomass of trees of species j in sample plot p of stratum i at a given point of time in year t ; t d.m.

$B_{TREE, l, j, p, t, i}$ = Biomass of tree l of species j in sample plot p of stratum i at a given point of time in year t ; t d.m.

A_i = Stratum i area; ha

$f_j(X_{1,l}, X_{2,l}, X_{3,l}, \dots)$ = Above-ground biomass of the tree returned by the allometric equation for species j relating the measurements of tree l to the above-ground biomass of the tree; t d.m.

R_j = Root-shoot ratio for tree species j ; dimensionless

A significant number of mangrove plants within the project area during the second monitoring period are in a transitional condition between saplings and trees. Field teams classified each mangrove plant as sapling or tree. For this second verification, in order to remain a conservative carbon analysis, saplings have not been accounted in the carbon monitoring.

Step 1: Selection of the allometric equation.

The following allometric equations have been selected and applied to the respective trees to calculate AGB per tree:

Table 13 Allometric equations used in AGB assessment

Applied to	Equation	R ²	N (sample trees used to derive the equation)	Source
Rhizophora spp.	$AGB = 0.235 * D^{2.42}$	0.98	57	Ong et al. (2004) ²⁰
Other mangrove trees	$AGB = 0.251 * \rho * (D)^{2.46}$	0.98	104	Komiyama et al. (2005) ²¹

Where:

AGB = Above ground biomass (kg)

ρ = Wood density, g/cm³

D = Diameter at Breast Height or Diameter at 30cm- or 10cm from the last stilt root that is fixed to the surface (d30 or d10) and from the ground (dbh) (cm)²². For Rhizophora spp. normally the d₃₀ or d₁₀ is taken. For Avicennia dbh is taken.

²⁰<http://www.sciencedirect.com/science/article/pii/S0378112703003906> (also provided as supporting documentation)

²¹Komiyama, A.; Pongparn, S. & Kato, S. (2005) Common Allometric equations for estimating the tree weight of mangroves. Journal of tropical Ecology 21: 471-477. Cambridge University Press (available as supporting documentation), also in http://www.cifor.org/publications/pdf_files/WPapers/WP86CIFOR.pdf

²² For stilt rooted species (e.g. Rhizophora spp.), stem diameter is often measured above the highest stilt root. For some individuals with prop roots extending well into the canopy, it is not necessary, practical or accurate to measure above the highest prop root. Typically, tree diameter is measured above the stilt roots, where a true main stem exists. In Rhizophora species the d₃₀ is normally recorded. If it is not possible to measure d₃₀ then d₁₀ is measured. For non stilt rooted species (e.g. Avicennia) with a main stem, dbh is taken)

Species specific values of wood density were applied in order improve calculations accuracy using a global wood density database²³. A genus average value has been applied for the species not listed in the database. A site average value was used when neither the species nor the genus was listed in the database. a table with different examples of mangrove species wood densities , with a classification based on four priorities:

1. Species from South-East Asia (tropical)
2. Species from South-East Asia
3. Genus from South-East Asia (tropical)
4. Genus from South-East Asia

Table 14 Wood density of project-relevant mangrove species (Zanne et al. (2011))

Scientific name	Wood density (gcm ⁻³)
Avicennia spp.	0.61
Rhizophora apiculata	0.85
Rhizophora mucronata	0.82
Rhizophora stylosa	0.84
Other	0.68

Step 2: Calculation of change in carbon stock in tree biomass

This step is developed in the following sub-steps:

- Sub-step 2.1: Calculation of above ground biomass (AGB) per tree ($f_j(X_{1,i}, X_{2,i}, X_{3,i}, \dots)$) based on the trees variables measured and using the selected allometric equation (see results in Excel file “2019_10_26_Yagasu_Ex post_Calculations_2nd verification”);
- Sub-step 2.2: Calculation of tree biomass ($B_{TREE,i,j,p,t,i}$) using equation 16, based on the results of sub-step 2.1. and using the selected default root-shoot ratio: 0.29;
- Sub-step 2.3: Calculation of the carbon stock in tree biomass within the project boundary at t_2 (C_{TREE,t_2} , second monitoring ending date) and t_1 (C_{TREE,t_1} , first monitoring ending date). C_{TREE,t_2} is calculated in the Excel file “2019_10_26_Yagasu_Ex post_Calculations_2nd verification” based on equations 9 to 15 (see summary results in table below), C_{TREE,t_1} results come from the previous verification report and monitoring report;

²³Zanne et al. (2009) Global wood density database.
<http://datadryad.org/resource/doi:10.5061/dryad.234/1>

- Sub-step 2.4: Discount rate application to the calculation of change in carbon stock in trees during the period between two points of time based on monitoring uncertainty. Calculations provide an uncertainty of 59% (calculated using equations 1 and 2 of AR-TOOL14).

Table 15 Summary results in the calculation of ΔC_{TREE}

Stratum ID	A_i (ha)	n_i	w_i	$b_{TREE,t,i}$ (t d.m./ha)
1	761.92	75	0.15	5.12
2	2,554.51	96	0.49	4.46
3	1,910.82	66	0.37	1.03
Total	5227.25	237	1.00	Weighted mean 10.05

A_{PLOT} (ha)	0.01
$b_{TREE,t}$ (t d.m./ha)	10.6
t_{val}	1.6512
u_c (%)	19.5%
CF	0.47
$B_{TREE,t}$ (t d.m.)	55,456.2
$C_{TREE,t}$ (t CO ₂ -e)	95,569.6

Table 13. Uncertainty of ΔC_{TREE}

	2nd monitoring (t2)	1st monitoring (t1)
A_{PLOT} (ha)	0.010	0.010
$b_{TREE,t}$ (t d.m./ha)	10.61	3.70
t_{val}	1.65	1.655
u_c (%)	19.5%	19.0%
CF	0.47	0.47
$B_{TREE,t}$ (t d.m.)	55,456.25	16839.70
$C_{TREE,t}$ (t CO ₂ -e)	95,569.6	29020.40
RE_{max} (%)	19.5%	19.0%
T (years)	3.83	
ΔC_{TREE}	66,549.2	
$u_{\Delta c}$ (%)	29%	
Discount (75% of $u_{\Delta c}$)	22%	
$\Delta C_{TREE,t}$ (t CO ₂ -e)	51,950.52	

Based on the results of above tables the conservative value (applying the discount rate) of ΔC_{TREE} has been calculated as 51,950.52t CO₂-e ($\Delta C_{TREE} / T = 51,950.52 / 3.8 = 13,563.81$ t CO₂-e). The discount has been applied following Appendix 2 of AR-TOOL14 "Applying uncertainty discount" where is it indicated that for uncertainties higher that 20% the discount rate should be 75% of the uncertainty.

Carbon stock change in shrubs

The project area was reforested with mangrove trees only. Due to the small size of overall vegetation in the project area, shrub vegetation is conservatively not estimated in this second verification. It may be included in subsequent verifications if it becomes significant.

Carbon stock change in dead wood

The change in carbon stock in dead wood is calculated using the conservative default-factor based method included in AR-TOOL12. The following equation is applied:

$$C_{DW,i,t} = C_{TREE,i,t} * DF_{DW} \quad \text{Equation 19 of this MR, 9 of AR-TOOL12}$$

Where:

- $C_{DW,i,t}$ = Carbon stock in dead wood in stratum i at a given point of time in year t ; t CO₂-e;
- $C_{TREE,i,t}$ = Carbon stock in trees biomass in stratum i at a point of time in year t ; t CO₂-e;
- DF_{DW} = Conservative default factor expressing carbon stock in dead wood as a percentage of carbon stock in tree biomass; per cent.

As stated in the PD, the conservative default factor for dead wood biomass for this project as influenced by the biome, elevation and annual precipitation is 6% of carbon stock in tree biomass. Hence, this value is applied. Therefore: $\Delta C_{DW} = \Delta C_{TREE} * DF_{DW} = 51,805.00 \text{ t CO}_2\text{-e} * 6\% = \mathbf{3,047.24 \text{ t CO}_2\text{-e}}$.

Estimation of changes in soil organic carbon stocks

Changes in carbon stocks in the SOC pool is calculated as indicated in the Methodology AR-AM0014:

$$\Delta SOC_{PROJ,t} = \frac{44}{12} * \sum_{t=1}^t A_{PLANT,t} * dSOC_t * 1year \quad \text{Equation 20 of this MR, 4 of AR-AM0014}$$

Where:

- $\Delta SOC_{PROJ,t}$ = Change in SOC stock within the project boundary, in year t ; t CO₂-e
- $A_{PLANT,t}$ = Area planted in year t , ha
- $dSOC_t$ = The rate of change in SOC stocks within the project boundary, in year t , tCha⁻¹yr⁻¹.

Calculation of dSOC has been done based on the publication Murdiyarso et al. (2015)²⁴, where the potential of Indonesian mangrove forest was analyzed, providing evidence that mangroves in Indonesia are among the ecosystems with the highest carbon accumulation in the world and that soil carbon accounts for close to 80% of all carbon pools in these systems. This study analyzed soil carbon in mangroves up to 200 cm of soil depth. In order to be conservative, calculations for this project only considers 50 cm. In addition, in order to account for allochthonous soil carbon buried at the project sites, the default approach of the recently accepted VCS Methodology VM0033 Methodology for Tidal Wetland and Seagrass Restoration is applied²⁵. A deduction factor for allochthonous soil carbon for each of the soil layers up to 50 cm measured in the Murdiyarso et al. (2015) study was derived with this approach.

Based on this study, dSOC was estimated as 3.32 t C/ha/year.

As indicated in the IPCC supplement for wetlands, where activity results in patchy or patches of biomass, the emission factor recommended as the default value can only be applied when the mangrove cover is at least 10% of the overall area. All areas in the project with less than this mangrove cover were excluded from the project, so all the area under plantation for the second monitoring period can be included in the SOC calculations with this default value. This methodology was applied to the areas of this project.

Table 16 Derivation of eligible project areas for SOC accounting

Year of inclusion of project area	Area under plantation (ha). First activity instance	Area under plantation (ha). Second activity instance
2011 (6 months)		7.74
2012		51.95
2013		12.73
2014		169.90
2015	4,949.02	1.34
2016	4,949.02	26.35
2017	4,949.02	8.21
2018	4,949.02	

²⁴ Murdiyaso et al. (2015). The potential of Indonesian mangrove forest. Nature Climate Change. Available as supporting documentation.

²⁵ <http://www.v-c-s.org/sites/v-c-s.org/files/VM0033%20Tidal%20Wetland%20and%20Seagrass%20Restoration%20v1.0%2020%20NOV%202015.pdf>. In particular, Section 8.1.4.3 of this new methodology and equations 32, 33, 34, 37, 77.

Year of inclusion of project area	Area under plantation (ha). First activity instance	Area under plantation (ha). Second activity instance
2019 (7 months)	4,949.02	
Total	4,949.02	278.23

For this monitoring period, a 3.83-year period is used to calculate the actual soil organic carbon stock changes of the project. In this verification period, for the first time, the second project activity instances are included as shown in the table below.

Table 17 Calculation of SOC change for this monitoring period

Year	Project year	Annual carbon stock change (tCO ₂ -e)	Cumulative carbon stock change (tCO ₂ -e)
2011 (6 months)	1	47.1	47.1
2012	2	410.1	457.2
2013	3	803.5	1,260.7
2014	4	1,914.1	3,174.8
2015	5	17,878.6	21,053.4
2016	6	63,319.3	84,372.7
2017	7	63,529.4	147,902.1
2018	8	63,579.4	211,481.5
2019 (7 months)	9	37,014.2	248,495.6
Total		248,495.64	

Actual net greenhouse gas removals by sinks - $\Delta C_{\text{ACTUAL},t}$

The actual net GHG removals by sinks for the different years of the monitoring period are summarized below, separately for trees (including dead wood) and soil organic carbon, as well as totals.

In addition, the cumulative months of project implementation are shown, starting with 6 months in 2011 and ending with 7 months in 2019 up to the end of this monitoring period (30-07-2019).

Table 18 Plantation areas included in this monitoring period

Year	Plantation areas of first project activity instance (ha)	Plantation areas of second project activity instance (ha)	Period included in this verification- 1st instance (months)	Period included in this second verification- 2nd instance (months)
2011	191.18	7.74	0	6
2012	851.32	51.95	0	6
2013	1,986.53	12.73	0	6
2014	1,623.19	169.90	0	6
2015	296.80	1.34	3	6
2016		26.35	12	6
2017		8.21	12	6
2018			12	12
2019			6.99	6.99
	4,949.02	278.23		

Table 19 Actual net greenhouse gas removals by sinks for the different years of the monitoring period

	Actual tree net GHG removals by sinks, in year t;	Actual DW net GHG removals by sinks, in year t;	Actual SOC net GHG removals by sinks, in year t;	Actual total net GHG removals by sinks, in year t; (tCO ₂ e)
Project year	$\Delta_{PLA,t}$	$\Delta_{DW,t}$	$\Delta_{SOC_PROJ,t}$	$\Delta_{PROJ,t}$
	(tCO ₂ -e)	(tCO ₂ -e)	(tCO ₂ -e)	(tCO ₂ -e)
2015	3,362.62	199.82	21,053.44	24,615.88
2016	13,563.81	800.86	63,319.26	77,683.93
2017	13,563.81	810.65	63,529.43	77,903.89
2018	13,563.81	771.41	63,579.35	77,914.58
2019	7,896.48	473.45	37,014.16	45,384.10
TOTAL	51,950.52	3,056.20	248,495.64	303,502.36

Other project emissions

GHG project emissions from burning of biomass do not occur in any of the mangrove systems in the project area.

4.3. Leakage

As demonstrated in Section 3.3 of the Project Description, leakage to due displacement of agricultural activities in the areas of this project can be accounted as zero.

Table 20 GHG emissions due to leakage, in year t

Project year	LK _t
	t CO ₂ -e
2015	0
2016	0
2017	0
2018	0
2019	0
Total	0

4.4. Net GHG Emission Reductions and Removals

The net anthropogenic GHG emission reductions and removals are calculated using equation 6 of the methodology AR-AM0014:

$$\Delta C_{AR-CDM,t} = \Delta C_{ACTUAL,t} - \Delta C_{BSL,t} - LK_t \quad \text{Equation 21 of this MR, 6 of AR-AM0014}$$

Where:

$\Delta C_{AR-CDM,t}$ = Net anthropogenic GHG removals by sinks, in year t; t CO₂-e

$\Delta C_{ACTUAL,t}$ = Actual net GHG removals by sinks, in year t; t CO₂-e

$\Delta C_{BSL,t}$ = Baseline net GHG removals by sinks, in year t; t CO₂-e

LK_t = GHG emissions due to leakage, in year t; t CO₂-e

The GHG emission reductions and removals for the first and second project activity instance are shown below.

Table 21 Total Yagasu project GHG emission reductions for this monitoring period

Year	Baseline emissions or removals $\Delta C_{BSL,t}$ (t CO ₂ -e)	Project emissions or removals $\Delta C_{ACTUAL,t}$ (t CO ₂ -e)	Leakage emissions LK_t (t CO ₂ -e)	Net GHG emission reductions or removals $\Delta C_{AR-CDM,t}$ (t CO ₂ -e)
2015	1,153.48	24,615.88	0	23,462.40
2016	124.74	77,683.93	0	77,559.19
2017	38.86	77,903.89	0	77,865.03
2018	0	77,914.58	0	77,914.58
2019	0	45,384.10	0	45,384.10
Total	1,317.08	303,502.36	0	302,185.29

Therefore, the total GHG emission reductions of the first and second project activity instances, for the second monitoring period from 01-10-2015 to 31-07-2019, are 302,185.29 t CO₂-e. The non-permanence risk rating is 10% – as assessed and documented in the AFOLU non-permanence risk report (provided as a separate document). Therefore, the total number of buffer credits that need to be deposited into the AFOLU pooled buffer account is 30,218.53t CO₂-e (10%). The number of buffer credits per vintage year of this monitoring period is:

Table 22 No of buffer credits per vintage year

Year	Buffer credits in year t; (tCO ₂ e)
2015	2,346.24
2016	7,755.92
2017	7,786.50
2018	7,791.46
2019	4,538.41
Total	30,218.53

The number of GHG credits eligible to be issued as VCUs for the first and second project activity instance of this monitoring period is 271,966.76t CO₂-e.